

Wireless Communication (EX751 / EX715)

Detailed PYQ

September 13, 2026

Notes

- PYQs from TU, IOE, BEX (EX751) and BEI (EX715) programmes combined, 2070–2082 BS. 23 papers: 16 BEX (2070–2080) and 7 BEI (2079–2082).
- BEX questions are in normal font; BEI questions are in **this monospace style**.
- Regular exam = **bold**, Back exam = regular font. “New Back (2066 & Later Batch)” counts as Back. Where a paper prints “Regular / Back” in one cell (**76 Bh**, **71 Bh**) it is taken as Regular.
- So: **72 Ash** = BEX Regular, 73 Ma = BEX Back, **79 Bh** = BEI Regular, 80 Ba = BEI Back.
- Months: Ba = Baishakh, Bh = Bhadra, Ash = Ashwin, Ma = Magh, Ch = Chaitra.
- Keywords: (I) = Importance/Need/Significance, (+) = Advantage, (-) = Disadvantage, +Fig = Figure, +Eg = Example, Vs = Compare/Differentiate.
- Source defects are recorded as printed, not corrected: 2073 Magh Q2 asks for cluster size, cluster count and maximum users but prints *no data*; the 2079 Chaitra header prints “BEI” on what is the EX751 (BEX) paper. Both are noted in place.
- Full verbatim transcriptions of all 22 papers: `ocr/EX-751_EX-715_OCR.md`.

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1 Introduction

(2 Hours)

1.1 Evolution of Wireless Communications (1G–5G)

1. Briefly explain evolution of wireless communications 1G–3G [6] (70 Ma, 72 Ma)
 - |→ Evolution of wireless comm. in terms of technology & worldwide market penetration [6] (70 Ma)
 - |→ Evolution of cellular radio 1G–3G [4] (**71 Bh**)
 - |→ Development of mobile comm., evolution path up to 3G technology [4] (74 Ma)
 - |→ Evolution of different generations of cellular systems [4] (**75 Bh**) [6] (82 Ba)
 - |→ Evolution from 1G to 2G, 2.5G in a TDMA-based cellular network [4] (71 Ma)
 - |→ Evolution of wireless comm. from second to third generation [4] (**73 Bh**)
 - |→ Evolution of wireless comm. from 1G to 4G, technology advancement [6] (**76 Bh**)
 - |→ Features introduced from second to fifth generation of cellular systems [4] (**78 Ch**)
2. Major benefits of wireless communication; brief description of the 4G wireless communication system [2+4] (**81 Bh**)

1.2 Comparison of Available Wireless Systems, Trends

1. 2G Vs 3G with +Eg of technologies used; prioritized handoff + cell dragging [4+2] (**72 Ash**)
2. 3G Vs 4G mobile comm. standards, technology advancement [4] (**75 Bh**)
 - |→ 1G Vs 2G Vs 3G Vs 4G standards, technology advancement [6] (**74 Bh**)
 - |→ 3G Vs 4G [4] (80 Ba, 81 Ba)
 - |→ 2G Vs 3G Vs 4G standards, technology advancement [4] (**80 Bh**)
 - |→ Improvements introduced in 2G, 3G and beyond-3G standards [6] (**70 Bh**)
 - |→ Features of 4G other than 3G mobile communication system [5] (**77 Ch**)

1.3 Trends in Cellular Radio Beyond 3G

1. Basic trends in wireless communication beyond 3G, focusing on key technological changes [4] (**82 Bh**)
2. Forward & reverse channel; 5G Vs 4G mobile communication [2+2] (**79 Bh, 79 Ch**)
 - |→ Define forward and reverse channel [2] (**80 Bh**)
 - |→ Forward & reverse channel; list the features of 5G [2+2] (**80 Ch**)

2 Cellular Mobile Communication Concepts

(4 Hours)

2.1 Frequency Reuse and Channel Assignment

1. Frequency reuse concept; co-channel reuse ratio in detail [6] (72 Ma)
|→ Minimizing reuse distance maximizes spectral efficiency (I) [4] (74 Bh)
|→ Cellular concept for $N = 7$ [3] (76 Bh)
2. Frequency reuse; various channel assignment strategies used in mobile communication [2+4] (82 Ba)
3. Prove co-channel reuse ratio $Q = \sqrt{3N}$, $N = i^2 + j^2$; find optimal N for omni-directional, 120° and 60° sectoring, SIR = 15 dB worst case, path loss exponent $n = 4$, considering trunking efficiency [4+4] (71 Bh)
|→ Prove $Q = \sqrt{3N}$; find optimal N for omni, “1200” (120°) and 60° sectoring, SIR = 15 dB, $n = 3$; comment on results [3+5] (79 Bh)
|→ Derive the formula for co-channel reuse ratio $Q = \sqrt{3N}$ [4] (80 Bh)
|→ Co-channel Vs adjacent channel interference; prove $Q = \sqrt{3N}$; 20 MHz total duplex spectrum, 25 kHz per simplex channel — find (a) number of duplex channels, (b) channels per cell for $N = 4$ [3+4+3] (70 Bh)
|→ Co-channel Vs adjacent channel interference; prove $Q = \sqrt{3N}$ [3+3] (81 Bh)
|→ Obtain $S/I = \sqrt[3]{3N}/i_0$ [4] (74 Ma)
4. For a seven-cell reuse pattern, find minimum distance between centres of co-channel cells; area of each cell is uniform and equal to 23 km^2 [4] (74 Bh)

2.2 Handoff Strategies, Types, Priorities, Practical Considerations

1. Define handoff margin +Fig [3] (75 Bh)
2. What is hand off? Handoff strategy used in GSM [8] (70 Ma)
3. Handover process in cellular system; types of handover with application [6] (74 Ma)
|→ Handover process; various practical handover considerations [4+4] (78 Ch)
|→ Practical handoff considerations [4] (79 Bh, 73 Ma)
|→ Prioritized handoff; cell dragging [2] (72 Ash)
|→ Cell dragging; how hand off is processed in a cellular system [2+4] (73 Bh)
|→ Improper handoff situation Vs proper handoff situation [4] (76 Bh)
|→ What is Hand-off? Proper and improper HO with necessary drawing [1+3] (81 Ba)
4. Two schemes for prioritizing handoff; define grade of service, traffic intensity, average holding time and block call [3+4] (82 Bh)

2.3 Interference and System Capacity

1. Explain the terminologies (i) co-channel interference (ii) adjacent interference [2.5+2.5] (80 Ch)
2. What is cell footprint? Explain interference tier in brief [1+4] (80 Ch)
3. Interference in wireless/mobile communication [4] (80 Ba, 81 Ba)

2.4 Grade of Service (GoS) and Trunking

1. GoS, and how it is measured in a “blocked call cleared” trunking system; cellular provider uses digital TDMA tolerating SIR = 15 dB worst case, path loss exponent $n = 3$ — find optimal N for (a) omni-directional antennas, (b) 120° sectoring, (c) 60° sectoring; comment [3+5] (79 Bh)

2. Urban area, population two million; System A has 394 cells with 19 channels each. Find the number of users supported at 2% blocking if each user averages two calls per hour at an average call duration of three minutes. Assuming trunk systems operate at maximum capacity, compute the percentage market penetration of the cellular provider

[5] (80 Ba, 81 Ba, **79 Ch**)

3. Mobile radio system: each user averages three calls per hour, each call lasting an average of 5 minutes. (a) traffic intensity per user; (b) number of users at 1% blocking with only one channel; (c) number of users at 1% blocking with five trunked channels; (d) if the users found in (c) are suddenly doubled, the new blocking probability of the five-channel trunked system — justify whether performance is acceptable

[2+2+2+3] (**77 Ch**)

Traffic (A) in erlangs for B%:

Trunks N	0.1%	0.2%	0.5%	1%	1.2%	3%	5%	7%
1	0.001	0.002	0.005	0.010	0.012	0.031	0.053	0.075
2	0.046	0.065	0.105	0.153	0.168	0.282	0.381	0.470
3	0.194	0.249	0.349	0.455	0.489	0.715	0.899	1.06
4	0.439	0.535	0.701	0.869	0.922	1.26	1.52	1.75
5	0.762	0.900	1.13	1.36	1.43	1.88	2.22	2.50

4. Cluster $N = 7$, blocking probability 1%, average call length two minutes, average request rate 1 call/hour per user — find the traffic capacity loss for a blocked-calls-cleared system due to trunking for 57 channels when going from omnidirectional antennas to 60° sectorized antennas

[2+2+2] (81 Ba)

GOS / Channel table as printed:

GOS / Channel	0.5%	1%	2%
1	.005	.0101	.0204
9	3.333	3.783	4.345
19	10.33	11.23	12.33
54	39.47	41.51	44.00
57	42.1	44.2	46.8

5. City of 1500 km² covered by a 12-cell system, each cell radius 1.387 km, total spectrum 28.5 MHz, full duplex channel bandwidth 25 MHz, GOS 0.02 for a blocked-calls-cleared system, offered traffic per user 0.03 Erlangs, traffic intensity per cell 84 Erlang — compute (a) number of cells in the service area, (b) channels per cell, (c) maximum carrier traffic, (d) total users served at 2% GOS, (e) mobiles per unique channel, (f) theoretical maximum users served at one time

[12] (72 Ma)

6. Cellular system with 32 cells, each of 1.6 km radius, reuse factor 7, supporting 336 traffic channels in total — determine the total geographical area covered, the number of traffic channels per cell and the total number of simultaneous calls supported

[4] (**82 Bh**)

7. Determine (a) the cell cluster size, (b) the number of cell clusters in the service area, (c) the maximum number of users in the service area at any instant

[6] (73 Ma)

The paper prints no data for this question — recorded as printed.

2.5 Coverage and Capacity Enhancement

1. Why do we need the microcell zone concept? Explain it in brief [3] (80 Ba, 81 Ba) [1+3] (**79 Ch**)
2. Telephone network company expanding capacity in a city: sectoring Vs cell splitting — as a cellular planning engineer, which option is best and why? [5] (**75 Bh**) [4] (**81 Bh**)
- How cell splitting and sectoring improve coverage and capacity [5] (**73 Bh**)

3. Techniques used for enhancing capacity & coverage in a cellular radio network [8] (71 Ma)
4. A single high-power transmitter supports 40 voice channels over an area of 140 km^2 with the available spectrum. The area is divided equally into seven smaller cells, each served by a lower-power transmitter so that each cell supports 30% of the total voice channels — find (i) the coverage area of each cell, (ii) the total number of voice channels available in the cellular system compared with the non-cellular system

[1+3] (82 Ba)

3 Radio Wave Propagation in Mobile Network Environment

(12 Hours)

3.1 Free Space Propagation, Radiated Power and Electric Field

1. Free space propagation: receiver 10 km from a 50 W transmitter, carrier 900 MHz, $G_T=1$, $G_R=2$, receiver antenna purely real $50\ \Omega$ and matched — find (i) power received, (ii) magnitude of E-field at the receiver antenna, (iii) power flux density, (iv) rms voltage applied to the receiver input
[8] (80 Ba, 81 Ba) [2+2+2+2] (78 Ch)
2. TX output 165 W at 325 MHz into a 12 dBi antenna; RX 15 km away with 6 dBi gain; find power delivered to the receiver (free-space, no other losses or mismatches)
[6] (80 Bh)
3. Link budget: BS transmitter 10 W at 250 MHz, 20 m of RF coax at 3 dB/100 m, antenna gain 9 dBi; receiving antenna 25 km away, gain 4 dBi; negligible feeder loss but 0.2 dB receiver mismatch loss ($50\ \Omega$ system) — find power delivered to the receiver assuming free-space propagation
[8] (72 Ash)
4. Determine the radio coverage range of a base station transmitting 150 W, receiver threshold -104 dBm, path loss at the first metre 15 dB, path loss exponent 4
[6] (70 Bh)
5. Mobile 5 km from a base station with a vertical $\lambda/4$ monopole of gain 2.55 dB; E-field at 1 km from the transmitter is 10^{-3} V/m, carrier 900 MHz — (a) length and effective aperture of the receiving antenna; (b) received power using the two-ray ground reflection model, TX height 50 m, RX height 1.5 m
[6] (71 Bh)
6. Cordless phone operating at 52000 kHz with a range of 50 m — determine the free space path loss
[6] (77 Ch) [2+4] (80 Ch)
7. Transmitter produces 50 W — express transmit power in (i) dBm and (ii) dBW; with 50 W into a unity-gain antenna at 900 MHz find the received power in dBm at a free-space distance of 100 m, and $P_r(10\text{ km})$, assuming unity receive gain
[1+1+2+2] (81 Ba)
8. Explain basic propagation models; compute the far-field distance for an antenna of maximum dimension 1 m operating at 900 MHz
[3+3] (79 Ch)

3.2 Large-Scale Path Loss: Propagation Mechanisms

1. Large scale Vs small scale propagation model; propagation mechanisms which have impact on propagation in a mobile environment
[3+6] (72 Ash)
|→ The three basic radio wave propagation mechanisms [3] (73 Bh)
|→ Mobile radio propagation in terms of large scale path loss and small scale fading [8] (72 Ma)
2. Define path loss; parameters of a mobile multipath channel used to classify the various types of fading
[2+4] (80 Bh)
3. Derive the expression for phase difference in the two-ray free space propagation model [8] (71 Bh)
|→ Derive path difference and phase difference between the LOS path and the ground-reflected (GR) path in the two-ray model using simple geometrical relations [6] (81 Bh)
|→ Diffraction in radio wave propagation; derive an expression for phase difference in the Fresnel Zone Geometry model [2+6] (79 Bh)
|→ (+) of the two-ray propagation model over the free space path loss model; derive the path loss equation for the two-ray model +Fig [2+6] (76 Bh)

4. What is known as scattering? Derive an expression for the two-ray ground reflected model [2+8] (74 Bh)
5. Path loss Vs fading of signal; time dispersion fading and its types [2+6] (71 Bh)

3.3 Practical Link Budget Design Using Path Loss Models

1. Explain the log-distance path loss model [4] (77 Ch)
2. Estimate the distance to be maintained on the reverse link between one BTS and a mobile, with an appropriate link budget diagram: (i) mobile antenna 20 dBi, TX power 15 dBm, receive sensitivity -75 dBm; (ii) BTS antenna 5 dBi, TX power 20 dBm, receive sensitivity -80 dBm; (iii) short cables, 3 dB loss each side, 900 MHz. Use Okumura's model for mean path loss with $G(\text{Area}) = 9$ dB, $A_{mu} = 43$ dB, $h_{re} = 10$ m, $h_{te} = 100$ m; link margin (reverse link) = 10 dB [10] (76 Bh)
3. Estimate the feasibility of a 10 km wireless link in a suburban area with one access point and one client radio, using the Okumura model for path loss; median attenuation 20 dB, gain due to environment 13 dB, AP antenna height 100 m, client antenna height 10 m: (a) AP antenna 5 dBi, TX 20 dBm, sensitivity -80 dBm; (b) client antenna 20 dBi, TX 15 dBm, sensitivity -75 dBm; (c) short cables, 3 dB loss each side, 2.4 GHz [12] (74 Bh)

3.4 Outdoor Propagation Models

1. Explain any two outdoor propagation models used in the mobile radio environment [3+3] (75 Bh, 71 Ma) [5] (73 Ma)
2. Necessary conditions for the Okumura and Hata models [2+2] (80 Ch, 79 Ch)
|→ Compare the Okumura model with the Hata model [4] (81 Bh)
3. Okumura model: median path loss for T–R separation 50 km, transmit antenna effective height 100 m, receive antenna effective height 10 m, suburban area, correction factor 9 dB at 900 MHz, median attenuation relative to free space 43 dB [4+4] (74 Ma)
|→ Okumura model, suburban, $d=50$ km, $h_{te}=100$ m, $h_{re}=10$ m, BS radiating EIRP of 1 kW at 900 MHz — find EIRP (dBm) and the power at the receiver where the receiving antenna gain is 10 dB, medium attenuation factor 43 dB, gain due to area 9 dB [4+2+2] (80 Bh) [5] (82 Bh)
|→ Okumura model, suburban, $d=50$ km, $h_{te}=100$ m, $h_{re}=10$ m, BS radiating EIRP of 1 kW at 900 MHz — find the median path loss and the power at the receiver for a unity-gain receiving antenna, $A_{mu} = 43$ dB, $G_{AREA} = 9$ dB [8] (82 Ba)
|→ Okumura model: compute the transmitter–receiver separation if median loss is 167 dB at carrier 2.1 GHz, transmitting antenna 40 m, receiving antenna 2 m, large city, $A_{mu} = 34$ dB, $G_{area} = 0$ dB [8] (77 Ch)
4. Hata model: correction factor and pathloss for a medium-size city, $f_c=950$ MHz, BS antenna 45 m, distance 10 km, MS antenna 5 m; compute free space pathloss and compare with Hata pathloss [4+4] (79 Bh)
|→ Hata model: maximum cell radius and corresponding cell area, BS transmitting 50 W, minimum acceptable received power at MS -91 dBm, carrier 900 MHz, $h_{BS}=30$ m, $h_{MS}=1$ m, medium sized city [6] (73 Ma)
|→ Hata model: propagation path loss for a 900 MHz cellular system in a large urban city, BS transmitter antenna 100 m, mobile receiver antenna 2 m, mobile at 4 km [7] (73 Bh)
Hints printed on the 73 Bh paper:

$$L_{50} = 69.55 + 26.16 \log f_c - 13.82 \log h_t - \alpha(h_r) + (44.9 - 6.55 \log h_t) \log d$$

$$\alpha(h_r) = (1.1 \log f_c - 0.7)h_r - (1.56 \log f_c - 0.8) \quad \text{small to medium city}$$

$$\alpha(h_r) = 8.29(\log 1.54h_r)^2 - 1.1 \quad \text{large city, } f_c \leq 300 \text{ MHz}$$

$$\alpha(h_r) = 3.2(\log 11.75h_r)^2 - 4.97 \quad \text{large city, } f_c > 300 \text{ MHz}$$

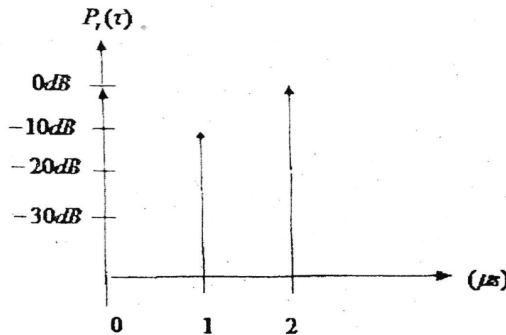
5. Two-ray mobile point-to-point propagation model: path loss at 800 MHz, transmitting antenna 30 m, receiving antenna 2 m, distance 10 km; compare with the free-space path loss model [4+4] (70 Ma)
6. What do you mean by the Ericsson Multiple Breakpoint model? [2] (80 Bh)

3.5 Indoor Propagation Models

1. Explain indoor propagation models (any two) [8] (70 Bh)
2. Factors that influence the indoor propagation model [5] (82 Bh)

3.6 Small Scale Fading, Multipath Parameters, Doppler

1. What is small scale fading? Types in radio propagation + factors which influence it [2+4+4] (75 Bh) [4+4] (80 Ba, 81 Ba) [2+2] (73 Ma) [3+6] (72 Ma)
 |→ Fading; factors influencing small-scale fading; derive the relation for Doppler's shift [1+2+5] (78 Ch)
 |→ Define fading; its different types based on delay spread and Doppler spread [1+3] (81 Ba)
2. Parameters of the mobile multipath channel [7] (71 Ma)
3. Define Doppler spread; derive the expression for Doppler shift [2+4] (81 Ba)
 |→ Significance of frequency dispersion in wireless communication; derive the relation for Doppler spread [2+4] (82 Bh)
 |→ Doppler's effect in small scale fading [2] (80 Ch)
4. Doppler spread; types of small-scale fading based on Doppler spread; mean excess delay and rms delay spread for the multipath profile below; 90% and 50% coherence bandwidth of the channel [4+4] (70 Ma)



5. Coherence bandwidth and coherence time — definitions and significance in mobile radio propagation [4] (73 Ma, 77 Ch) [2+5] (82 Bh)
 |→ Coherence bandwidth and coherence time, and their significance in mobile radio propagation; fading and its various types [3+5] (82 Ba)
 |→ Doppler spread and coherence bandwidth; classify fading on the basis of rms delay spread and coherence time [4+4] (74 Ma)
 |→ Frequency selective fading; relationship between coherence bandwidth and delay spread in the time domain [2+7] (79 Ch)
6. Determine the smallest symbol period T_s , and thus the greatest symbol rate, that can be sent through an RF channel with the power delay profile below without using an equalizer; modulation gives suitable BER whenever $\sigma_\tau/T_s \leq 0.1$ [8] (75 Bh)

Power [dB]	0	0	-10	-20
Delay [μ s]	0	50	75	100

7. Mean excess delay, rms delay spread and maximum excess delay for the multipath profile $P_r(\tau) = (-20 \text{ dB}, -10 \text{ dB}, -10 \text{ dB}, 0 \text{ dB})$ at $\tau = (0, 1, 2, 5)$ second; also estimate the 50% coherence bandwidth [6] (**81 Bh**)
8. rms delay spread and 90% coherence bandwidth of the channel below; is the channel flat fading or frequency selective for (i) an AMPS system of transmission bandwidth 30 kHz, (ii) a GSM system of transmission bandwidth 200 kHz? [6] (71 Ma)

Power [dB]	0	-10	-20	-23
Delays [ns]	0	100	200	400

3.7 Types of Small-Scale Fading; Rayleigh and Ricean Distribution

1. Rayleigh fading channel Vs Rician fading channel, with appropriate expressions [2] (71 Ma)
 $\rightarrow$ Rayleigh and Ricean fading distribution [5] (70 Ma, **72 Ash**) [5] (**73 Bh**)
2. What is frequency selective fading? Explain the Rayleigh distribution in brief [2+4] (**80 Ch**)

4 Modulation–Demodulation Methods in Mobile Communications

(4 Hours)

4.1 Digital Linear Modulation: BPSK, DPSK, QPSKs

1. QPSK: transmission & detection process [6] (75 Bh) [8] (72 Ma)
|→ QPSK transmission & detection + Fig, constellation diagram [8] (81 Bh)
|→ QPSK modulation with its appropriate equation and constellation diagram [7] (73 Bh)
2. OQPSK transmitter & receiver; why $\pi/4$ -QPSK is preferred over OQPSK [5+2] (71 Bh)
3. DPSK modulation: transmitter and receiver; pseudo-noise (PN) sequence and why it is used [4+2+2] (72 Ma)
4. BPSK against QPSK modulation [5] (73 Ma)

4.2 Constant Envelope Modulation: MSK, GMSK

1. MSK as a special type of continuous-phase FSK and OQPSK; compare OQPSK, MSK and GMSK in bandwidth efficiency and power efficiency; which is suitable for GSM modulation and why [3+3+2] (80 Ba, 81 Ba)
|→ What is MSK modulation? [2] (80 Ch)
|→ MSK and GMSK modulation techniques [4] (76 Bh)
2. Advantages of digital modulation over analog; GMSK transmission & detection process with block diagram and constellation diagram [2+6] (79 Bh) [8] (81 Ba)
3. MSK & GMSK modulation techniques; OFDM modulator and demodulator + Fig [8] (70 Ma)

4.3 M-ary Modulation (MPSK, MFSK, QAM) and OFDM

1. OFDM operation + Fig (block diagram of transmitter & receiver) [8] (74 Bh, 80 Bh, 78 Ch, 82 Ba)
|→ OFDM: generalize the modulation and demodulation technique [8] (71 Ma)
|→ OFDM principle; different types of spread spectrum modulation techniques [4+4] (70 Bh)
|→ OFDM transmitter in brief [4] (79 Ch)
|→ OFDM receiver in brief [4] (80 Ch)
|→ OFDM transmitting side block diagram; (+) and (-) [4+4] (82 Bh)
2. Why do we need M-ary QAM? Compare 16-QAM and 16-PSK using their constellation diagrams [2+6] (77 Ch)

4.4 Spread Spectrum: PN Sequences, DS and FH

1. Direct Sequence & Frequency Hopped Spread Spectrum techniques [4] (73 Ma)
|→ Define spread spectrum modulation; DSSS with block diagrams of transmitter and receiver [2+6] (82 Bh) [6] (81 Ba) [2+5] (80 Ch)
|→ Define spread spectrum; FHSS with relevant diagram [2+5] (79 Ch)
|→ FHSS operation with a neat block diagram [4] (73 Bh)
2. Advantages of spread spectrum modulation technique in wireless communication [4] (76 Bh)
|→ Major disadvantages of the spread spectrum system [2] (77 Ch)

5 Equalization and Diversity Techniques

(4 Hours)

5.1 Equalization: Linear, Non-linear, Adaptive

1. Equalization (I) in wireless communication; describe the signal processing operation that minimizes the effects of ISI [2+6] (**75 Bh**) [2+5] (72 Ma) [4] (73 Ma)
|→ Signal processing operation that minimizes ISI; RACK receiver; structure of the linear transversal equalizer +Fig [2+1+4] (**82 Bh**)
|→ Signal processing operation that minimizes ISI; RAKE receiver; working of an M-branch RAKE receiver [2+2+4] (82 Ba)
2. Training and tracking modes of operation for adaptive equalizers [4] (**78 Ch**)
|→ Why equalization is needed; training and tracking modes in detail [1+3] (**73 Bh**)
3. MSE algorithm: determine the optimal solution of the weights, with appropriate diagram and derivation [8] (**79 Bh**) [7] (**80 Ch**, **79 Ch**)
4. Maximum Likelihood Sequence Estimation (MLSE) +Fig; define time diversity and explain two implementations of it [6+1+5] (**72 Ash**)
|→ MLSE equalization +Fig [4] (71 Ma, 72 Ma)
|→ Working of MLSE [4] (**81 Bh**)
5. Adaptive equalization algorithms (any two) [4] (73 Ma)
|→ Adaptive Equalization [5] (74 Ma)
6. Interleaving: working principle +Fig [4] (**81 Bh**)
|→ Why interleaving is needed in wireless communication; its working in brief [4] (**80 Bh**)
|→ Define interleaving; how equalization offsets the ISI introduced by the multipath time-dispersive channel, using an adaptive equalizer as example [2+6] (81 Ba)

5.2 Diversity Methods and Combining

1. Why equalization & diversity are needed in wireless communication systems; diversity reception methods used for space diversity; explain any one [3+3+2] (80 Ba, 81 Ba)
2. Why is there a need to implement diversity? Maximum ratio combining space diversity technique [2+2] (**80 Bh**)
|→ Diversity (I) in wireless communication; different types of diversity techniques [2+6] (**71 Bh**)
|→ Diversity techniques; combining methods used for space diversity [4+4] (74 Ma)
|→ Why implement diversity? Various diversity combining techniques [4+6] (**74 Bh**)
|→ Various space diversity techniques along with block diagrams [4] (73 Ma)
|→ What is diversity? Any two types of diversity techniques in detail [2+6] (**70 Bh**)
|→ How diversity improves the quality of the network [2] (**76 Bh**)
3. Discuss and compare different types of antenna diversity technique [4] (71 Ma)
|→ Feedback Vs maximum ratio combining technique used in antenna diversity [4] (**78 Ch**)

5.3 RAKE Receivers and Interleaving

1. RACK (RAKE) receiver: working of an M-branch receiver [8] (**72 Ash**) [3+5] (**77 Ch**)
|→ RAKE receiver: operation with appropriate block diagram [4] (70 Ma, **76 Bh**)
|→ What is a RAKE receiver and how does it exploit the concept of time diversity? [3] (**73 Bh**)
|→ RAKE receiver [4] (79 Bh)

6 Speech and Channel Coding Fundamentals

(4 Hours)

6.1 Speech Characteristics, Frequency Domain Coding, Vocoders

1. Characteristics of the speech signal [8] (**72 Ash**) [6] (**77 Ch**) [3] (73 Ma, 79 Bh) [3] (**80 Ch**)
|→ Characteristics of speech signals; how they are used in designing coders [8] (**72 Ash**) [4] (**82 Bh**, 81 Ba)
2. Vocoder (I); explain any two predictive coders [2+6] (**70 Bh**)
|→ Vocoder with block diagram; different kinds of vocoders [2+3] (73 Ma) [3+5] (**81 Bh**)
|→ Operation of a vocoder with the help of a block diagram [4] (**73 Bh**) [6] (81 Ba)
|→ Formant vocoder operation; characteristics of the speech signal [4+4] (**71 Bh**)
|→ Characteristics of speech signals; basic working principle of the Formant and Channel vocoder [4+4] (**82 Ba**)
|→ Why we need speech coding techniques; basic concept of VOCODER [4+4] (70 Ma) [2+6] (**78 Ch**)
|→ Waveform and voice coding techniques; characteristics of speech; GSM CODEC +Fig [4+2+2] (74 Ma)
|→ Operation of any two source coders used in speech coding [6] (**74 Bh**)
|→ Types of frequency domain coding of speech [4] (**73 Bh**)
3. Characteristics of the speech signal; Linear Predictive Coder (LPC) with neat block diagram [2+6] (80 Ba, 81 Ba, 72 Ma) [3+5] (**79 Bh**, 73 Ma, **76 Bh**) [2+6] (**75 Bh**, 72 Ma)
|→ Types of linear predictive coder [6] (71 Ma)
4. Sub-band coding with transmitter and receiver [6] (**80 Bh**)

6.2 Channel Coding: Block, Convolutional, Turbo

1. What is channel coding? [2] (71 Ma)
2. 7-bit Hamming code method with a relevant example [5] (**80 Ch**)
3. Convolutional encoding and decoding [5] (**74 Bh**)
4. Viterbi decoding algorithm [3] (**75 Bh**, 70 Ma, **72 Ash**) [5] (**73 Bh**)
5. Define turbo coding [2] (**82 Bh**) [5] (74 Ma)

7 Multiple Access in Wireless Communications

(9 Hours)

7.1 FDMA

1. Non-linear effect in FDMA; total spectrum allocation 25 MHz, guard band at the edge of the spectrum 100 kHz, channel bandwidth 200 kHz — find the number of channels available in the FDMA system [3+2] (80 Ba, 81 Ba)
2. Number of radio channels in an FDMA system: a US analog mobile phone system is allocated 12.8 MHz for every simplex band, total spectrum allocated 12.8 MHz, guard bandwidth 10 kHz, channel bandwidth 30 kHz [4] (79 Ch, 77 Ch)
|→ Total bandwidth 12.5 MHz, guard bandwidth 10 kHz, channel bandwidth 30 kHz — find the number of channels available in an FDMA system [4] (78 Ch)
3. FDMA (+) and (-) [4] (73 Ma)

7.2 TDMA

1. TDMA (+) over FDMA [4] (71 Ma, 73 Ma)
2. GSM time slot of 6 trailing bits, 8.25 guard bits, 26 training bits and 2 traffic bursts of 58 bits of data — find the frame efficiency [4] (70 Bh)
3. N-TDMA system using a one-way bandwidth of 25 MHz for the forward (or reverse) channel, system bandwidth divided into 30 kHz radio channels, guard bands $B_g = 20$ kHz — calculate the number of simultaneously transmitting users that can be accommodated in each cluster [4] (80 Ch)

7.3 Spread Spectrum Multiple Access: CDMA, FHMA, Hybrids

1. CDMA: characteristics, (+) and limitations [2+6] (79 Bh)
|→ Important features of CDMA in wireless communication [2] (80 Ch)
|→ CDMA design considerations [4] (80 Ba, 81 Ba)
|→ CDMA (+) and (-) [4] (73 Ma)
|→ (+) of the CDMA cellular system over the TDMA cellular system [4] (73 Bh)
2. Implementation of CDMA with encoding and decoding parts, with relevant example [7] (82 Bh)
[2+6] (77 Ch) [8] (79 Ch)
3. Construct an 8 by 8 Hadamard code / Hadamard matrix H_8 (Walsh codes for CDMA) [4] (80 Bh)
[3] (82 Bh)
|→ Steps to satisfy the condition for a Hadamard code; construct the 8 by 8 Hadamard code satisfying them [3+5] (79 Ch)
4. SSMA: DS & FH spread spectrum techniques; (+) (-) of FDMA, TDMA and CDMA [4+4] (73 Ma)
|→ Different types of spread spectrum multiple access techniques; FDMA Vs CDMA [6+2] (72 Ma)
|→ Spread Spectrum Multiple Access, SSMA variants and application +Fig [8] (74 Ma)
5. FHMA working with +Eg [4] (79 Bh)
|→ FHMA technique with block diagram (transmitter & receiver) [6] (80 Bh) [7] (80 Ba, 81 Ba) [4] (78 Ch)
|→ FHMA principle; two hybrid spectrum multiple access techniques that mitigate the near-far problem [4+6] (74 Bh)
|→ FHMA [5] (75 Bh, 76 Bh) [4] (73 Ma)
|→ FHMA operation; types of FHMA [4+2] (81 Bh)
|→ (+) of CDMA over TDMA; Frequency Hopping Spread Spectrum (FHSS) multiple access technique +Fig; different types of FHSS [4+6+2]

(82 Ba)

6. Explain any two hybrid multiple access techniques [4] (80 Bh) [6] (80 Ch)
|→ Two types of hybrid spread spectrum multiple access techniques with (+) and (-) [6] (81 Bh)
|→ Self-jamming problem in CDMA; FHMA operation + Fig; any two hybrid spread spectrum multiple access techniques with (+) and (-) [2+4+6] (72 Ash)
|→ FHMA principle; near-far effect in CDMA and how it is solved [3+4] (71 Ma)
|→ Define near-far effect; any one hybrid SSMA technique that mitigates it [2+2] (71 Ma)
|→ Multiple access; near-far effect in CDMA and all possible mitigation techniques [5+4] (76 Bh)

7.4 SDMA

1. What is space division multiple access? Any two hybrid spread spectrum multiple access techniques which minimise the near-far effect [2+6] (75 Bh)

7.5 Multiple Access Comparison

1. Define multiple access; explain TDMA, CDMA and SDMA [2+6] (71 Bh)
|→ Multiple access; Time Division CDMA (TCDMA) and Time Division Frequency Hopping [7] (71 Bh)
|→ What is a multiple access technique? Compare FDMA with CDMA [2+4] (70 Ma)
2. FDMA Vs CDMA Vs TDMA: (+) and (-) of each [4] (73 Ma)
|→ Compare and contrast TDMA and FDMA [4] (81 Ba, 78 Ch)
3. Merits and demerits of Code Division Multiple Access [6] (70 Bh)
4. What do you mean by duplexing? Explain its different types [2+4] (81 Ba)

7.6 Wireless LAN Standards

1. Wireless Local Area Network (WLAN) [3] (75 Bh, 72 Ash)
2. WiFi [5] (76 Bh)

8 Wireless Systems and Standards

(6 Hours)

8.1 GSM: Architecture and Channels

1. GSM architecture: operation of each component [8] (80 Ba, 81 Ba)
|→ GSM architecture: draw & explain [8] (70 Ma, **79 Bh**, 73 Ma) [4+4] (71 Ma)
|→ GSM System Architecture [5] (**73 Bh**)
|→ Components of the Network Switching Subsystem in GSM architecture [4] (**73 Bh**, **82 Bh**)
2. GSM traffic & control channels [8] (**75 Bh**, **72 Ash**, **78 Ch**)
|→ Broadcast Control Channel (BCH) and Common Control Channel (CCCH) in GSM [4+4] (**80 Bh**)
|→ Broadcast Control Channel of the GSM system [5] (**79 Ch**)
|→ Operation of the different control channels used in the GSM system [8] (**81 Bh**)
|→ Types of control channels available in GSM and their purposes [2+4] (82 Ba)
3. GSM frame structure; show that TDMA frame efficiency cannot reach 100% in GSM [6+2] (74 Ma)
|→ GSM frame structure; how traffic and control channels are used while making a call [4+4] (**70 Bh**) [4+6] (**77 Ch**)
|→ Frame structure of GSM [5] (**76 Bh**)
|→ Channel structure of GSM [6] (74 Ma)
|→ GSM frame hierarchy [4] (**82 Bh**)
4. Basic signal processing operations to convert a speech signal into a radio signal and back in GSM [8] (**71 Bh**, 81 Ba)
5. Specifications of GSM [5] (**74 Bh**)

8.2 CDMA Standards: IS-95 Forward and Reverse Channels

1. Channelization code; forward channels in cdma IS-95 [4] (73 Ma)
|→ Draw the forward CDMA (IS-95) channel [4] (**80 Bh**) [4] (**78 Ch**) [6] (**80 Ch**, 82 Ba)
|→ Draw the reverse CDMA (IS-95) channel [4] (81 Ba)
|→ Pilot and sync channels in the IS-95 forward link +Fig [6] (**73 Bh**)
2. GSM Vs CDMA standards [4] (71 Ma) [3] (**76 Bh**)
3. CDMA Vs LTE system architecture; functions of the entities [3+5] (74 Ma)

8.3 WiFi, WiMAX, LTE and Recent Trends

1. WiMAX [3] (**72 Ash**, 70 Ma) [5] (73 Ma) [4] (**78 Ch**)
2. LTE [3] (**72 Ash**)

8.4 Regulatory Issues

1. Regulatory issues in wireless communication [4] (80 Ba, 81 Ba) [5] (**74 Bh**, **76 Bh**)
|→ Spectrum regulations [4] (79 Bh)
|→ Regulatory issues related to spectral licensing [5] (**73 Bh**)
|→ Regulatory issues related to licensing and spectrum allocation [4] (**81 Bh**)
2. Significance of spectrum management; functions of the regulatory [2+4] (**80 Ch**) [2+3] (**79 Ch**)